

Multiple Choice (10 marks)

1. The pressure at the bottom of a pond does NOT depend on the
 - (a) salinity of the water.
 - (b) proximity to the equator.
 - (c) presence of fish in the pond.
 - (d) type of pond vegetation.
 - (e) acceleration due to gravity.
2. How many megacoulombs of positive charge are in 1.00 mol of neutral molecular-hydrogen gas (H_2) ?
 - (a) 0.75 MC
 - (b) 0.90 MC
 - (c) 0.10MC
 - (d) 1.00 MC
 - (e) 0.50MC
3. By locating the north celestial pole (NCP) in the sky how can you determine your latitude?
 - (a) The altitude of the NCP is your angular distance from the North Pole.
 - (b) The azimuth of the NCP is your angular distance from the equator.
 - (c) The altitude of the NCP is your angular distance from the equator.
 - (d) The altitude of the NCP is the same as your latitude.
 - (e) The altitude of the NCP is the same as your longitude.
4. A microwave oven is connected to an outlet, 120 V, and draws a current of 2 amps. At what rate is energy being used by the microwave oven?
 - (a) 240 W
 - (b) 120 W
 - (c) 10 W
 - (d) 480 W
 - (e) 360 W
5. Using the Bohr formula, calculate the wave numbers of the first five members of the spectrum emitted by ionized helium.

- (a) 9845.56, 15200.17, 18450.43, 20550.25, 22000.88
- (b) 10243.17, 15478.43, 18977.56, 20878.25, 22317.88
- (c) 9950.21, 15100.32, 18350.43, 20450.54, 21950.65
- (d) 9877.56, 15243.17, 18478.43, 20578.25, 22017.88
- (e) 9600.12, 14800.34, 18000.56, 20100.78, 21500.90

True / False (10 marks)

Answer True or False

- 6. True or False: The speed of a wave cannot be determined if the wavelength is less than the frequency value in Hertz.
- 7. True or False: The surface of Venus is approximately 750 million years old, indicating a relatively young geological history.
- 8. True or False: The rotational inertia of an object is minimized when most of its mass is located near the rotational axis.
- 9. True or False: If the Sun were to collapse into a black hole, Earth would continue in its present orbit due to the conservation of angular momentum.
- 10. True or False: A 1000-gram metal block falling 10 meters converts approximately 98 calories into heat energy upon impact.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the process of calculating the frequency of the hydrogen transition from $n = 101$ to $n = 100$, including the relevant quantum mechanics principles and equations involved.
- 12. Discuss the implications of a scale reading less than a person's usual weight while standing in an elevator. What can this indicate about the elevator's motion, and how does circular motion affect perceived weight?

End of Paper